

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2025 4WM2HR Worked Solutions

May2025 | 4WM2HR | Unit 2

31 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 5****TOPIC: Statistics Toolkit**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 1** The table shows information about the weights, in kilograms, of 50 dogs.

Weight (w kg)	Frequency
$4 < w \leq 8$	5
$8 < w \leq 12$	17
$12 < w \leq 16$	15
$16 < w \leq 20$	9
$20 < w \leq 24$	4

- (a) Write down the modal class.

..... kg
(1)

- (b) Work out an estimate for the mean weight of the 50 dogs.

..... kg
(4)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 5

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

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- 2 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

3 Solve the simultaneous equations

$$3x + 2y = 10$$

$$3x - 4y = 16$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 3 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Simultaneous Equations**

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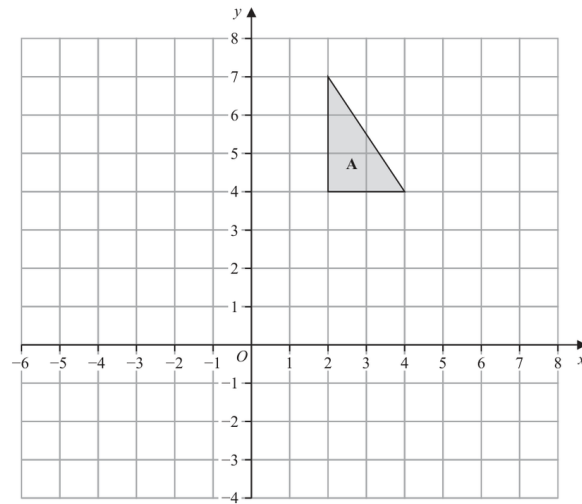
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PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Transformations**

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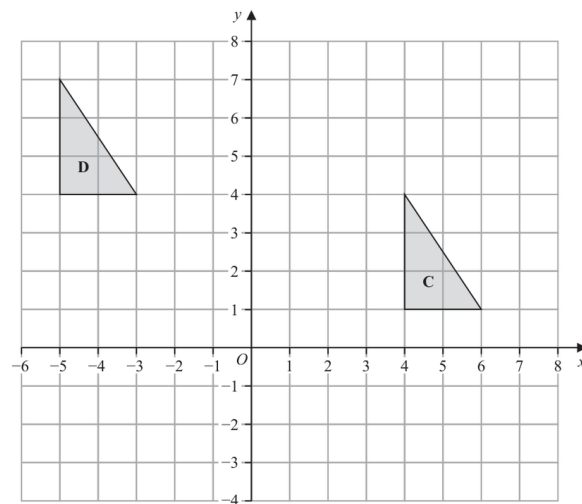
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- (a) On the grid above, rotate triangle A 90° anticlockwise about $(1, 2)$

Label your triangle B

(2)



- (b) Describe fully the single transformation that maps triangle C onto triangle D

(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Transformations**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Percentages**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

- 5** Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 m^2

Joshua buys one box of tiles for each 1.5 m^2 of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

..... %

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 5

TOPIC: **Percentages**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

6 (a) Write 82 000 000 in standard form.

.....
(1)

(b) Write 5.8×10^{-6} as an ordinary number.

.....
(1)

(c) Write 160×10^{300} in standard form.

.....
(1)

(Total for Question 6 is 3 marks)

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PAST PAPER SOLUTION

Q#: 6

Marks: 3

TOPIC: **Powers, Roots & Standard Form**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Statistics Toolkit**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

7 The mean of 7 numbers is 60

The mean of 3 of the numbers is 46

Work out the mean of the other 4 numbers.

(Total for Question 7 is 3 marks)

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PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Statistics Toolkit**

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PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Percentages**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

8 In a sale, normal prices are reduced by 15%

The sale price of a dishwasher is 612 Swiss francs.

Work out the normal price of the dishwasher.

..... Swiss francs

(Total for Question 8 is 3 marks)

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PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: **Percentages**

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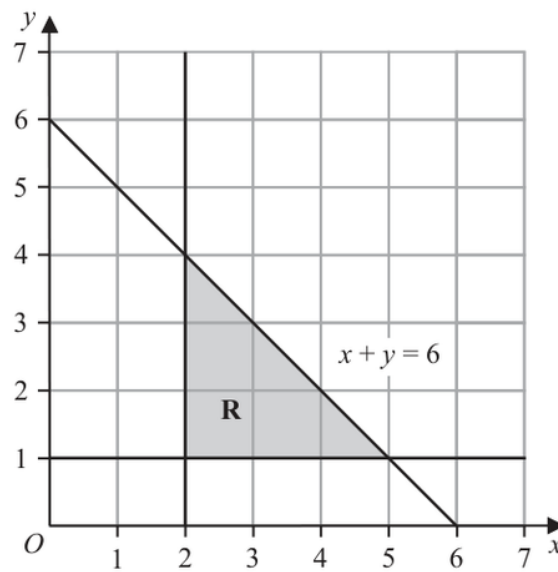
PAST PAPER QUESTION**Q#: 9****Marks: 6****TOPIC: Graphing Inequalities**

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- 9 (a) Solve the inequality $3x - 22 > 8x - 40$
Show clear algebraic working.

.....
(3)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....
(3)

(Total for Question 9 is 6 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 6

TOPIC: **Graphing Inequalities**

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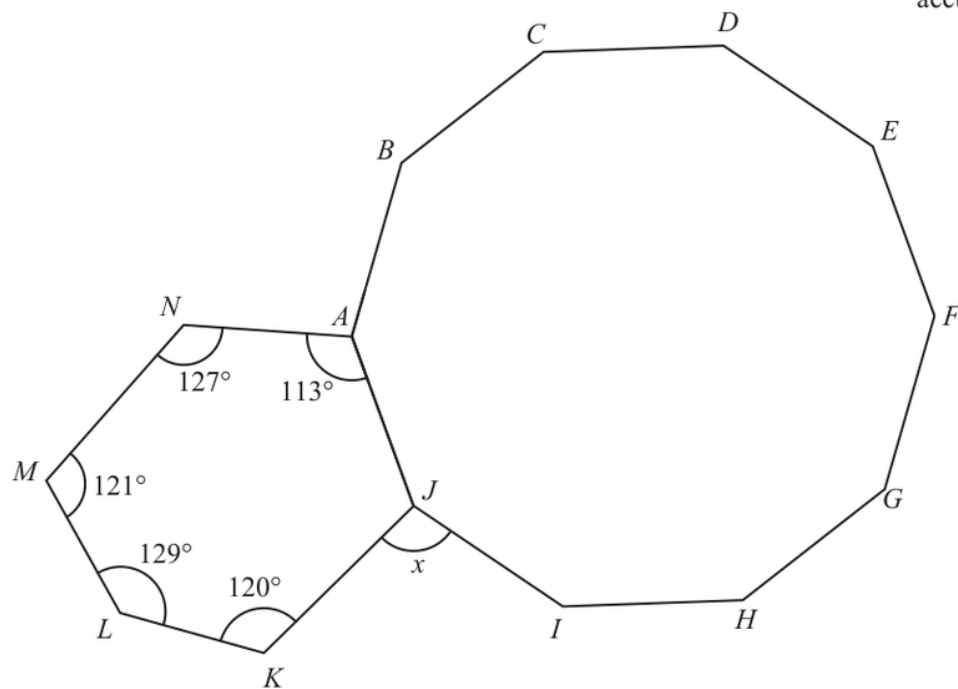
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PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

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- 10** The diagram shows a regular 10-sided polygon $ABCDEFGHIJ$ and a hexagon $AJKLMN$

Diagram **NOT**
accurately drawn



Work out the size of the angle marked x
Show your working clearly.

(Total for Question 10 is 5 marks)

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Q#: 10

Marks: 5

TOPIC: **Angles in Polygons & Parallel Lines**

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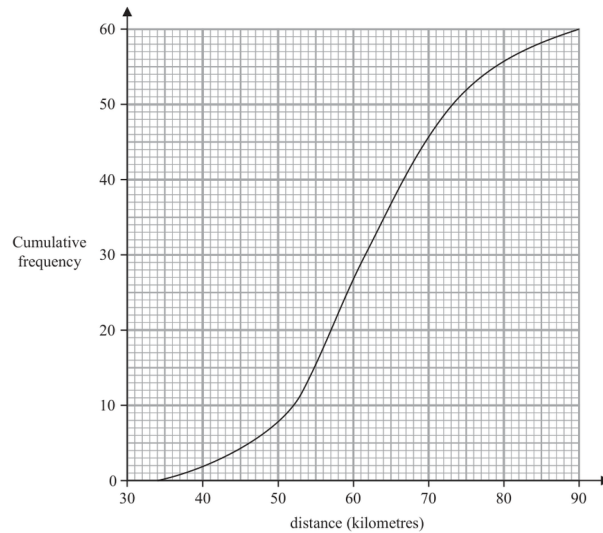
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PAST PAPER QUESTION**Q#: 11****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

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- 11 The cumulative frequency graph shows information about the distance, in kilometres, Ehrick drives each day for 60 days.



- (a) Use the graph to find an estimate for the median of the distances.

..... kilometres
(1)

- (b) Use the graph to find an estimate for the interquartile range of the distances.

..... kilometres
(2)

- (c) Work out the percentage of days that Ehrick drives for more than 75 kilometres. Give your answer correct to one decimal place. Show your working clearly.

..... %
(3)

Tomas and Alison record the number of kilometres that they drive each day for 60 days. The table gives information about their results.

	Interquartile range
Tomas	28 kilometres
Alison	15 kilometres

- (d) Which of Tomas or Alison is more consistent in the number of kilometres that they drive. Give a reason for your answer.

.....
(1)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Compound Interest & Depreciation**

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12 Osman buys a car for \$16 000

The car depreciates at a rate of 12% each year for the first 2 years.
In the third year, the car depreciates at a rate of $x\%$

At the end of 3 years, the value of the car is \$11 461.12

Work out the value of x

(Total for Question 12 is 3 marks)

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PAST PAPER SOLUTION

Q#: 12

Marks: 3

TOPIC: **Compound Interest & Depreciation**

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PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

13 The distance from Earth to Andromeda Galaxy is 1.6×10^{11} astronomical units.

- 1 astronomical unit = 1.5×10^8 km
- 1 light year = 9.5×10^{12} km

Work out the distance, in light years, from Earth to Andromeda Galaxy.
Give your answer in standard form, correct to 2 significant figures.

..... light years

(Total for Question 13 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Powers, Roots & Standard Form**

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PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Rearranging Formulas**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

14 Make k the subject of $p = \frac{8k^2 + 5}{7 - 3k^2}$

(Total for Question 14 is 4 marks)

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PAST PAPER SOLUTION

Q#: 14

Marks: 4

TOPIC: **Rearranging Formulas**

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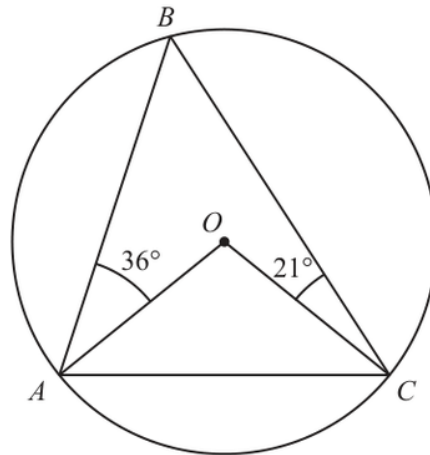
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PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Circle Theorems**

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15 A , B and C are points on a circle, centre O Diagram **NOT**
accurately drawnAngle $BAO = 36^\circ$ Angle $BCO = 21^\circ$ Work out the size of angle ACO

(Total for Question 15 is 3 marks)

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Q#: 15

Marks: 3

TOPIC: **Circle Theorems**

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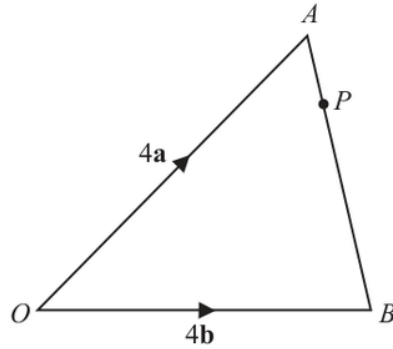
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PAST PAPER QUESTION**Q#: 16****Marks: 3**TOPIC: **Vectors**

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16 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 4\mathbf{a}$$

$$\vec{OB} = 4\mathbf{b}$$

P is the point on AB such that $AP : PB = 1 : 3$

(a) Write down $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....
(1)

(b) Express $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

.....
(2)

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 3

TOPIC: **Vectors**

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PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

17 The function g is defined as

$$g(x) = \frac{3x}{x-2}$$

(a) State the value of x that cannot be included in any domain of g
(1)(b) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$$g^{-1}(x) = \dots$$

(3)

(Total for Question 17 is 4 marks)

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PAST PAPER SOLUTION

Q#: 17

Marks: 4

TOPIC: **Functions**

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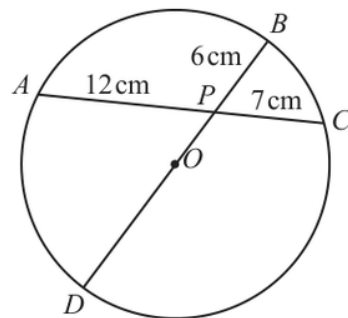
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PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Circle Theorems**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

18Diagram **NOT**
accurately drawn A, B, C and D are points on a circle, centre O DB is a diameter of the circle. AC and DB intersect at the point P

$$AP = 12 \text{ cm} \quad PC = 7 \text{ cm} \quad PB = 6 \text{ cm}$$

Work out the area of the circle.

Give your answer correct to 3 significant figures.

..... cm^2 **(Total for Question 18 is 3 marks)**

PAST PAPER SOLUTION

Q#: 18

Marks: 3

TOPIC: **Circle Theorems**

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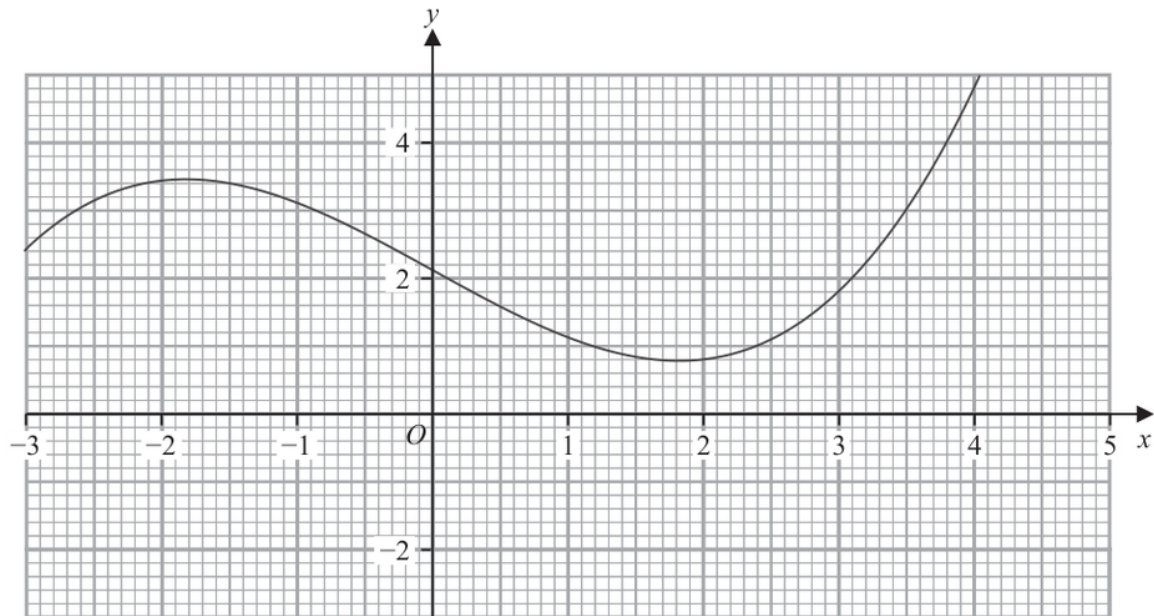
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PAST PAPER QUESTION**Q#: 19** **Marks: 3****TOPIC: Estimating Gradients**

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19 The diagram shows the graph of $y = f(x)$



Find an estimate for the gradient of the graph at the point where $x = 3$
Show your working clearly.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19

Marks: 3

TOPIC: **Estimating Gradients**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

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21 The diagram shows a solid cuboid.

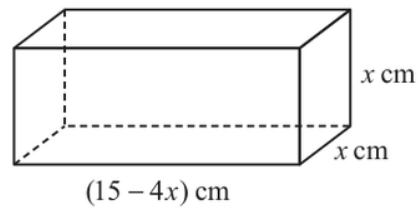


Diagram **NOT**
accurately drawn

The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

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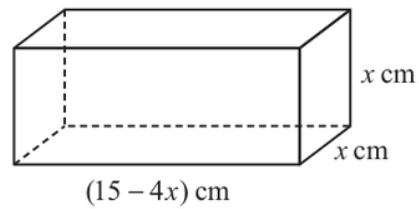
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PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

21 The diagram shows a solid cuboid.Diagram **NOT**
accurately drawnThe volume of the cuboid is $V \text{ cm}^3$ Find the maximum value of V

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

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22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

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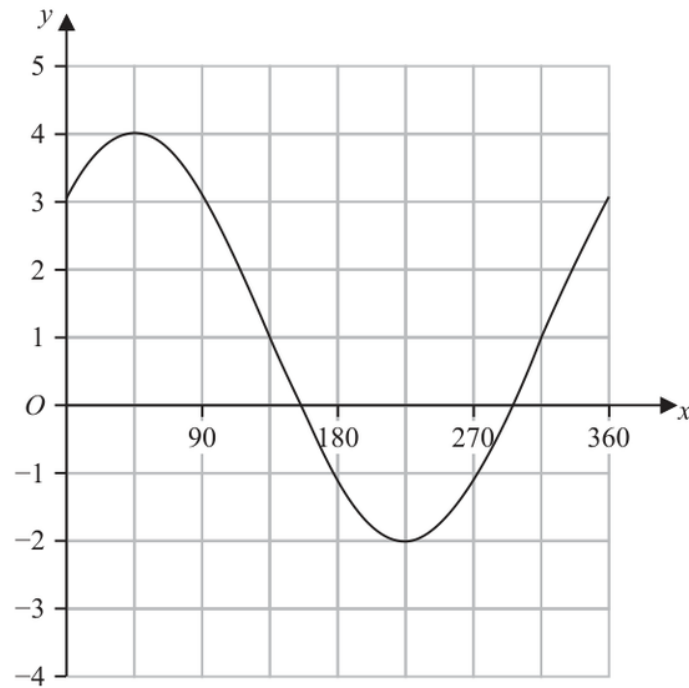
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PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Graphs of Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

23 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Graphs of Functions**

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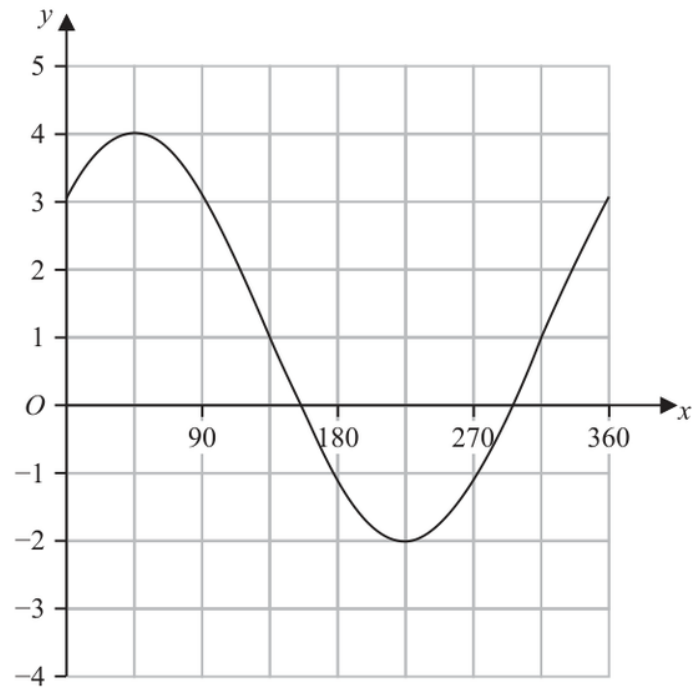
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PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Graphs of Functions**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

23 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Graphs of Functions**

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PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Sequences**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

24 An arithmetic series P has first term a and common difference d

The sum of the first 40 terms of P is 2070

An arithmetic series Q has first term $2a$ and common difference $3d$

The 25th term of Q is 186

Work out the sum of the first 40 terms of Q

Show clear algebraic working.

(Total for Question 24 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Sequences**

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PAST PAPER QUESTION**Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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An arithmetic series Q has first term $2a$ and common difference $3d$

The 25th term of Q is 186

Work out the sum of the first 40 terms of Q

Show clear algebraic working.

(Total for Question 24 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Sequences**

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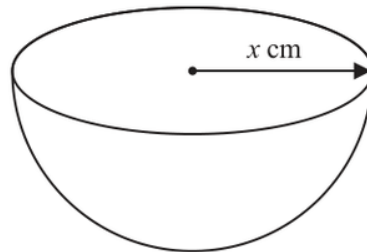
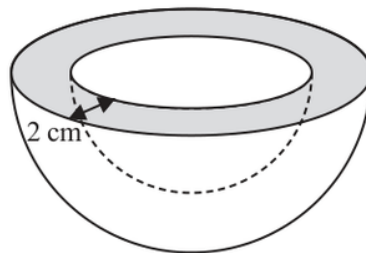
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PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Volume & Surface Area**

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25 The diagram shows a solid hemisphere, H Diagram **NOT**
accurately drawnThe radius of H is x cmThe volume of H is $6174\pi \text{ cm}^3$ A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cmDiagram **NOT**
accurately drawnWork out the **total** surface area of the bowl.
Give your answer in terms of π

..... cm^2

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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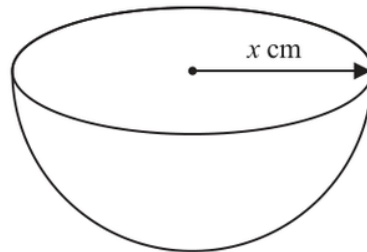
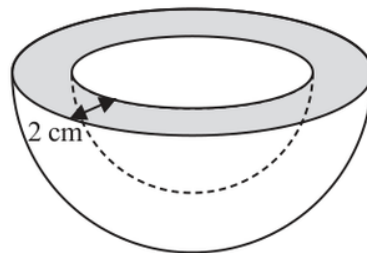
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PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Volume & Surface Area**

May 2025 4WM2HR - May2025 | 4WM2HR | Unit 2

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accurately drawnWork out the **total** surface area of the bowl.
Give your answer in terms of π

..... cm²

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Volume & Surface Area**

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